



Python for Fixed Income

COURSE 2 · BOND MATH

Spread over the Curve

WEEK 3 · DAY 2

Created by **Jeff Lenamon**

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Where does 147 come from?

5.606

Bezane Finance's yield, percent

147

its oas column, basis points

Three steps for every bond

1. Years to maturity, on 30/360
2. The curve yield at those years
3. The difference, times 100: basis points

A spread here describes the data. It is never a view on a bond.

Week 3: curve, spread, and the book

Day 11 Treasury curve interpolation

Day 12 Spread over the curve

Day 13 Spread duration and DTS

Day 14 Portfolio and scenarios

Day 15 The capstone

What the spread measures

$$\text{spread_bp} = (\text{yield_pct} - \text{curve yield at maturity}) \times 100$$

Bezane Finance, matures 2032-06-18	Percent
Yield	5.606
minus oas, 147 bp	1.47
Curve yield the file used	4.136

One percent is 100 basis points, so the gap in percent is multiplied by 100.

The invented curve, around it

Tenor, years	Yield, percent
3	3.916
5	4.086
7	4.214
10	4.348

4.136 sits between the 5-year and 7-year points.

Years to maturity on 30/360, the file's rule

Count	Days	Years
Actual, over 365	2088	5.720548
30/360, over 360	2058	5.716667

The file read the curve at `days_30_360 / 360: years_to_maturity`.

Bezane Finance, 30 September 2026 to 18 June 2032. Close, not equal: only one reproduces the file.

Excel side by side (tested in Excel)

<code>=YEARFRAC(D2,E2,0)</code>	5.716667
<code>=YEARFRAC(D2,E2,0)*360</code>	2058
<code>=YEARFRAC(DATE(2027,2,28),DATE(2031,8,31),0)*360</code>	1621
<code>=DAYS360(DATE(2027,2,28),DATE(2031,8,31))</code>	1620

YEARFRAC equals `years_to_maturity` on all 200 bonds. DAYS360 turns that 31st into a 30th: a lookalike, not the file's count.

The curve yield at each bond's maturity

```
# 5.7167 years: between the 5 and 7 year points
share = (years - 5) / (7 - 5)
curve = 4.086 + share * (4.214 - 4.086)
bondmath.interpolate_yield(years, tenors, yields)
```

Excel side by side (tested in Excel)

<code>=MATCH(G2,\$A\$2:\$A\$11,1)</code>	6: the 5-year point
<code>INDEX(...):INDEX(...)</code>	the two points
<code>=FORECAST.LINEAR(G2, yields, tenors)</code>	4.131867

The full formula is in the notebook. It has no flat ends: #N/A below the first tenor, #REF! at or above the last.

4.1319

the straight line, percent

4.136

what the file used, percent

The line gives 0.41 bp less. Section 12.6 explains why.

spread_to_curve_bp

```
def spread_to_curve_bp(yield_pct, years,
                       curve_tenors, curve_yields_pct):
    curve_yield_pct = interpolate_yield(years,
                                       curve_tenors, curve_yields_pct)
    # percent, times 100 basis points per percent
    return (yield_pct - curve_yield_pct) * 100
```

147.41

Bezane Finance to the ten-tenor table, bp

147

the oas column, bp

Excel side by side (tested in Excel)

$=(F2-H2)*100$ 147.4133

F2 is the yield in percent, as in the file. The *100 turns percent into basis points, as in the function.

The name says the unit: **_bp**.

The oas column, reproduced exactly

```
# the rule the file used (DATA.md)
yield_pct = round(3.55 + 1.05 x (1 - exp(-T / 7)), 3)
             + oas / 100
```

```
# a curve of one point: the bond's own years
spread_to_curve_bp(y, years, [years], [curve])
```

4.136

the formula at 5.7167 years, rounded

2.3e-13 bp

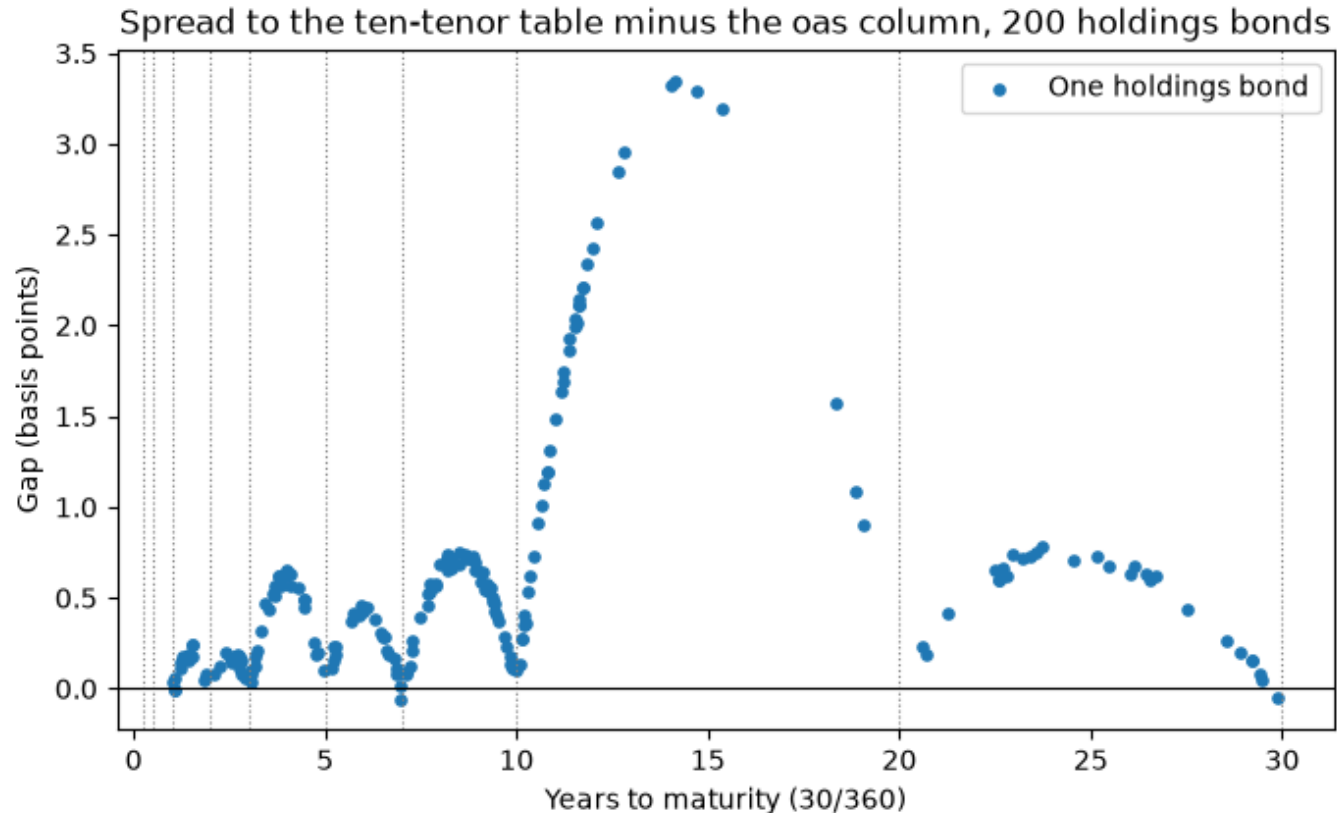
largest miss from oas, 200 bonds

Excel side by side (tested in Excel)

<code>=ROUND(3.55+1.05*(1-EXP(-G2/7)),3)</code>	4.136
<code>=(F2-J2)*100</code>	147.0000

Column K equals the oas column on all 200 bonds.

The interpolation choice moves the spread



3.35 bp

largest gap

0.64 bp

average gap

196 of 200: the table's spread is larger

The curve bends. A straight line between two points of it sits below it, so the curve yield is lower and the spread larger.

Compare two spreads only when they were built the same way.

What to call the number

oas

The file's column name. Here, built as a yield spread at the bond's own maturity.

spread_to_curve_bp

This course: one yield minus another, times 100. A yield spread.

OAS and Z-spread

On a desk: from a spot curve, each cash flow at its own rate. Beyond this course.

For a bullet with no options, an OAS comes out close to the Z-spread. The course's function is never called oas.

Today's Excel: one sheet, every formula tested

Col	Job	Excel, row 2 (tested)	Bezane Finance
G	Years	=YEARFRAC(D2,E2,0)	5.716667
H	Curve, straight line	FORECAST.LINEAR, INDEX/MATCH	4.131867
I	Spread to table, bp	=(F2-H2)*100	147.4133
J	Curve, the formula	=ROUND(3.55+1.05*(1-EXP(-G2/7)),3)	4.136
K	Spread to formula, bp	=(F2-J2)*100	147.0000
L	30/360 days	=YEARFRAC(D2,E2,0)*360	2058
M	The lookalike	=DAYS360(D2,E2)	2058

Curve in A1:B11; settlement in D, maturity in E, yield in percent in F. Fill row 2 down.

How the tests grew

`git log --oneline -- test_bondmath.py`

```
905c285 Add years_to_maturity and spread_to_curve...
f1906a0 Day 11: interpolate_yield on a curve
a298556 End of week 2: dates, accrued, price, dura...
5baded0 End of week 1: price and yield on a coupon...
```

With `--` and a file name, `git log` lists only the commits that changed that file.

`git diff --stat week2`

```
bondmath.py      | 57 +++
test_bondmath.py | 66 +++
2 files changed, 123 insertions(+)
```

With a tag, `git diff --stat` counts the lines changed since that commit: days 11 and 12.

The work folder is new each day, so the notebook rebuilds the week1 and week2 tags. Your own folder has them.

Bring it home: paste, test, commit, look back

1

Your folder

Set-Location to my_bondmath, in your projects folder.

2

Paste

Two functions and three tests at the end of your files, without the %%writefile line.

3

Test

python -m pytest -q: 35 passed, or more with your exercise functions.

4

Commit

git diff --stat, git add, git commit with a subject and a body.

5

Look back

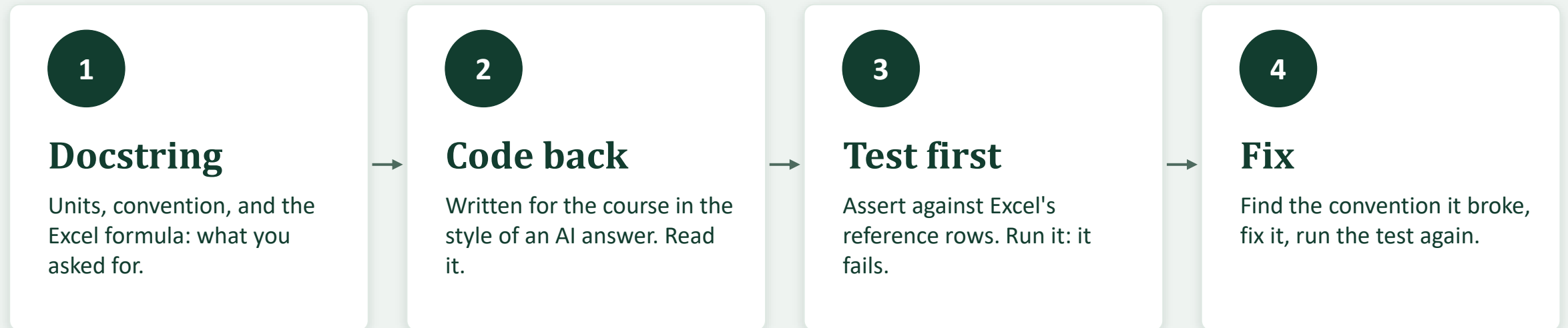
git log --oneline --test_bondmath.py, then git diff --stat week2.

unknown revision? `git log --oneline`, then `git tag week2 <day 10 hash>`

In Colab: download bondmath.py and test_bondmath.py from the work folder. They replace yesterday's downloads.

The AI check

The docstring is the prompt. The code that comes back runs, and has one convention error.



Today: the spread comes back in percent, and the code calls it basis points: 100 times too small.

Code that runs is not the same as code that is right.

What you can do now

- Say what a spread to the curve measures, in basis points
- Count years on 30/360 with `years_to_maturity`, or YEARFRAC basis 0
- Read the curve with `interpolate_yield`, or FORECAST.LINEAR and INDEX/MATCH
- Reproduce the oas column exactly; measure the interpolation gap
- Name it a yield spread; see growth with `git log -- file` and `diff --stat`

Week 3, Day 3 Spread duration and DTS: how far a price moves when the spread moves